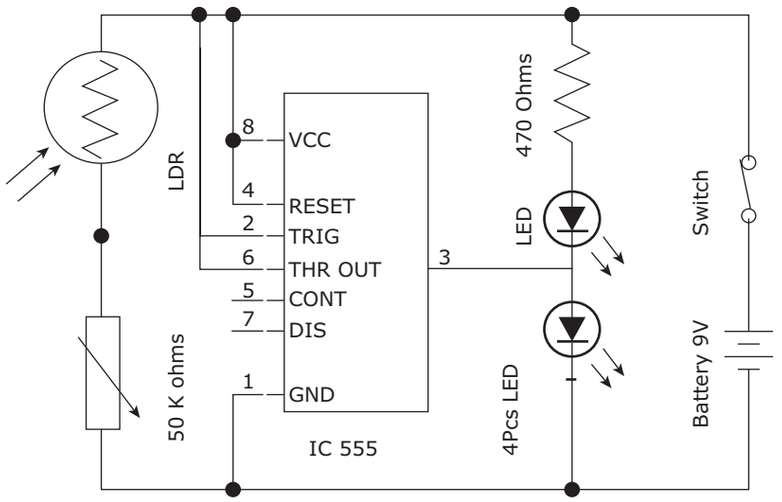
Activities

Construct Automatic Street light using LED and LDR



When an external voltage source (V) is connected to the photo-conductor, current I flows in the external circuit. If strong radiation incident on the LDR, the photo current may be as high as 10 mA, otherwise the LDR photo current may be negligibly small.

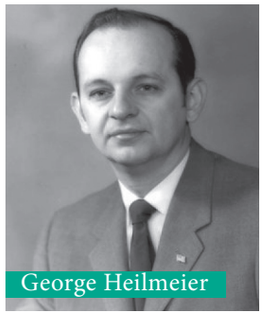
Applications

1. Used to measure intensity of light.
2. Used in counting system.
3. Used in burglar alarm.
4. Widely used in cameras to control shutter opening for flash light.
5. Used in automatic street light.

6.4. LIQUID CRYSTAL DISPLAY (LCD)

Liquid Crystal Display (LCD) are used for displaying of numeric and alphanumeric character in dot matrix and segmental displays. There are two types of liquid crystal materials used, they are nematic and cholesteric. In the Nematic Liquid Crystal (NLC), all the molecules align themselves approximately parallel to a unique axis, while retaining the complete translational freedom.

The liquid is normally transparent, but if subjected to a strong electric field, disruption of the well-ordered crystal structure takes place causing the liquid to polarise and turn opaque. The removal of the disruption of the crystal structure regains its original form and the materials become transparent.


George Heilmeier

History: Liquid crystal was first observed by an Austrian Botanist, Friedrich Reintzer in the year 1888. The material discovered by him was Cholesteryl Benzoate. In 1968, George Heilmeier and his team of scientists revealed the LCD to the public. Based on the construction, LCDs are classified into two types. They are,

1. Dynamic scattering type
2. Field effect type

6.4.1. Dynamic Scattering LCD

Construction: Figure 6.4(a) shows the construction of dynamic scattering LCD. The display consists of two glass plates, each coated with tin oxide (SnO_2) on the

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